Proposed change [A93], the arrivals of Appendix C follow the refill rule of the code

13 November 2026, version 1

How to read this

Part 1 says what is wrong. Part 2 shows Appendix C's "The arrivals" as it would read, and the notation rows. Green is new text, ~~struck~~ text goes, a removed equation is grey and marked "removed". Part 3 has the questions. Nothing is changed in the thesis yet.

Part 1. What is wrong

Appendix C defines how much arrives by the *top-up* rule. Each SKU is filled up to $(1 + \varepsilon)$ times last period's orders, never below a floor $x_{\min}$. The code has that rule (`stock_rule: topup`), but the thesis runs use the other one, `rule: refill` with `refill_units: 10` in every `configs/step12/fleet_<m>.yaml`.

Under the refill rule each SKU counts the units ordered since its last refill, from the moment a line is released (`multi_agent_env.py`, `_ordered_since_refill`). At an arrival it gets one refill of Q units for every Q it has counted, and the rest carries over (`storage/arrival.py`, `refill_amounts`). Every refill is one put-away to one shelf. Counting the units on their way and leaving out the units lines wait for, a SKU's stock then stays between its starting stock less Q and its starting stock.

Appendix G ("Setting") and the Problem Formulation already describe refills. Only Appendix C and two rows of the notation table still describe the top-up rule. $\hat{\rho}_t^{\text{unit}}$ is used only there, so its row goes too.

Part 2. The text

Appendix C, "The arrivals"

How much arrives is set by the orders alone, so it is the same under every rule. Stock comes back a refill at a time. Each SKU counts the units ordered since its last refill, from the moment a line is released, whether the units are picked yet or still wait. At a storage update a SKU gets one refill of Q units for every Q it has counted, and what is left over counts towards the next one. So with $Q = 10$, if 23 units of tea were ordered since the last refill, two refills of tea arrive and 3 units carry over. A SKU's stock, counting the units on their way and leaving out the units lines wait for, then stays between its starting stock less Q and its starting stock. This is the (R, nQ) policy of inventory control (Axsäter, 2015, `axsater2015inventory`), with the reorder

point R one refill below the starting stock. ~~Each SKU is filled up to what the last period's orders suggest, with a margin, and never to fewer than a floor of units. The units that lines still wait for are added on top. Arrivals count units, while $\hat{\rho}_t$ counts lines, so a second estimate counts the units ordered. So if 30 units of tea were ordered in the last period, the margin is 1 and the floor is 8, tea is filled up to 60 units plus the units still waited for. Units still on their way to a shelf count as stock, so stock already ordered is not ordered again, and the arrivals stay the same under every rule.~~

Formally, write $y_t^{\mathrm{ord}}(k)$ for the units of SKU k ordered since its last refill, just before the update at t, with "ord" for ordered. The *arrivals* are ~~Formally, with margin $\varepsilon \geq 0$, floor $x_{\min}$, $\hat{\rho}_t^{\mathrm{unit}}(k)$ the units of k ordered per step in the window, and $q_t^{\mathrm{wait}}(k)$ the units of k that lines wait for at t, the *arrivals* are~~

$$y_t(k) = Q \left\lfloor \frac{y_t^{\mathrm{ord}}(k)}{Q} \right\rfloor, \qquad \texttt{(eq:arrivals)}$$

and $y_t^{\mathrm{ord}}(k) - y_t(k)$, fewer than Q units, carries over.

Removed: $y_t(k) = \max\bigl(0,\ x_t^{\mathrm{up}}(k) + q_t^{\mathrm{wait}}(k) - \sum_v (x_t^{-}(k,v) - r_t(k,v) + x_t^{\mathrm{in},-}(k,v))\bigr)$, $x_t^{\mathrm{up}}(k) = \max(x_{\min}, \lceil (1+\varepsilon)\,\Delta\,\hat{\rho}_t^{\mathrm{unit}}(k) \rceil)$, ~~where $\Delta\,\hat{\rho}_t^{\mathrm{unit}}(k)$ is the number of units of k ordered in the period.~~

Notation table (Appendix A)

symbol	name	meaning
~~$\varepsilon \geq 0$~~	~~stock margin~~	~~Stock kept above the last period's orders~~
~~$x_{\min}$~~	~~stock floor~~	~~Fewest units a SKU is filled up to~~
~~$\hat{\rho}_t^{\mathrm{unit}}(k)$~~	~~unit demand estimate~~	~~Units of SKU k ordered per step . . .~~
$y_t^{\mathrm{ord}}(k)$	units ordered	Units of SKU k ordered since its last refill, before the update at t

The row for Q, the refill size, already exists. The same rows change in the unused copy `Chapters/2.Notation.tex`*, which follows Appendix A.*

Part 3. Questions

1. **The citation.** The code calls the rule the (R, nQ) policy. I could not open Axsäter's book to check its definition there. Keep the citation and the clause on the reorder point if it matches, or say only "ordering in fixed batches of Q" without a source?

2. **The worked numbers.** 23 units ordered, two refills, 3 carry over. Keep the example, or leave it to the equation?

3. **The stock bound.** The sentence that a SKU's stock stays between its starting stock less Q and its starting stock comes from the code's comment. It explains why a home shelf always has room (Appendix G, "Shelf capacity"). Keep it here, or only in Appendix G?